

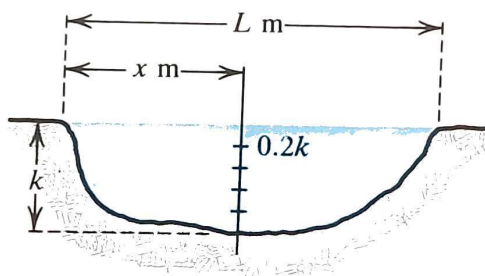
depth of the stream at a point x meters from the bank and $v(y)$ is the velocity (in m/sec) at a depth of y meters (see figure), then

$$\bar{v}_x = \frac{1}{k} \int_0^k v(y) dy$$

(see Definition (5.29)). With the *six-point method*, velocity readings are taken at the surface; at depths $0.2k$, $0.4k$, $0.6k$, and $0.8k$; and near the river bottom. The trapezoidal rule is then used to estimate $\bar{v}_x$. Given the data in the following table, estimate $\bar{v}_x$.

y (m)	0	$0.2k$	$0.4k$	$0.6k$	$0.8k$	k
$v(y)$ (m/sec)	0.28	0.23	0.19	0.17	0.13	0.02

EXERCISE 31



- c** 32 Refer to Exercise 31. The streamflow F (in m^3/sec) can be approximated using the formula

$$F = \int_0^L \bar{v}_x h(x) dx,$$

where $h(x)$ is the depth of the stream at a distance x meters from the bank and L is the length of the section. Given the data in the following table, use Simpson's rule to estimate F .

x (m)	0	3	6	9	12
$h(x)$ (m)	0	0.51	0.73	1.61	2.11
$\bar{v}_x$ (m/sec)	0	0.09	0.18	0.21	0.36

x (m)	15	18	21	24
$h(x)$ (m)	2.02	1.53	0.64	0
$\bar{v}_x$ (m/sec)	0.32	0.19	0.11	0

- c** Exer. 33–34: Use Simpson's rule with $n = 8$ to approximate the average value of f on the given interval.

33 $f(x) = \frac{1}{x^4 + 1}$; $[0, 4]$

34 $f(x) = \sqrt{\cos x}$; $[-1, 1]$

- c** Exer. 35–36: If f is determined by the given differential equation and initial condition $f(0)$, approximate f using the trapezoidal rule with $n = 10$.

35 $f'(x) = \frac{\sqrt{x}}{x^2 + 1}$; $f(0) = 1$

36 $f'(x) = \sqrt{\tan x}$; $f(0) = 2$

5.8 REVIEW EXERCISES

Exer. 1–42: Evaluate.

- 1 $\int \frac{8x^2 - 4x + 5}{x^4} dx$
- 2 $\int (3x^5 + 2x^3 - x) dx$
- 3 $\int 100 dx$
- 4 $\int x^{3/5} (2x - \sqrt{x}) dx$
- 5 $\int (2x + 1)^7 dx$
- 6 $\int \sqrt[3]{5x + 1} dx$
- 7 $\int (1 - 2x^2)^3 x dx$
- 8 $\int \frac{(1 + \sqrt{x})^2}{\sqrt[3]{x}} dx$
- 9 $\int \frac{1}{\sqrt{x}(1 + \sqrt{x})^2} dx$
- 10 $\int (x^2 + 4)^2 dx$
- 11 $\int (3 - 2x - 5x^3) dx$
- 12 $\int (x + x^{-1})^2 dx$
- 13 $\int (4x + 1)(4x^2 + 2x - 7)^2 dx$
- 14 $\int \frac{\sqrt[4]{1 - (1/x)}}{x^2} dx$
- 15 $\int (2x^{-3} - 3x^2) dx$
- 16 $\int (x^{3/2} + x^{-3/2}) dx$
- 17 $\int_0^1 \sqrt[3]{8x^7} dx$
- 18 $\int_1^2 \frac{x^2 - x - 6}{x + 2} dx$
- 19 $\int_0^1 \frac{x^2}{(1 + x^3)^2} dx$
- 20 $\int_1^9 \sqrt{2x + 7} dx$
- 21 $\int_1^2 \frac{x + 1}{\sqrt{x^2 + 2x}} dx$
- 22 $\int_1^2 \frac{x^2 + 2}{x^2} dx$
- 23 $\int_0^2 x^2 \sqrt{x^3 + 1} dx$
- 24 $\int_1^1 3x^2 \sqrt{x^3 + x} dx$
- 25 $\int_0^1 (2x - 3)(5x + 1) dx$
- 26 $\int_{-1}^1 (x^2 + 1)^2 dx$
- 27 $\int_0^4 \sqrt{3x}(\sqrt{x} + \sqrt{3}) dx$
- 28 $\int_{-1}^1 (x + 1)(x + 2)(x + 3) dx$
- 29 $\int \sin(3 - 5x) dx$
- 30 $\int x^2 \cos(2x^3) dx$
- 31 $\int \cos 3x \sin^4 3x dx$
- 32 $\int \frac{\sin(1/x)}{x^2} dx$
- 33 $\int \frac{\cos 3x}{\sin^3 3x} dx$

$$34 \int (3 \cos 2\pi t - 5 \sin 4\pi t) dt$$

$$35 \int_0^{\pi/2} \cos x \sqrt{3 + 5 \sin x} dx$$

$$36 \int_{-\pi/4}^0 (\sin x + \cos x)^2 dx$$

$$37 \int_0^{\pi/4} \sin 2x \cos^2 2x dx$$

$$38 \int_{\pi/6}^{\pi/4} (\sec x + \tan x)(1 - \sin x) dx$$

$$39 \int D_x \sqrt{x^4 + 2x^2 + 1} dx$$

$$40 \int_0^{\pi/2} D_x (x \sin^3 x) dx$$

$$41 D_x \int_0^1 (x^3 + x^2 - 7)^5 dx$$

$$42 D_x \int_0^x (t^2 + 1)^{10} dt$$

Exer. 43–44: Solve the differential equation subject to the given conditions.

$$43 \frac{d^2 y}{dx^2} = 6x - 4; \quad y = 4 \text{ and } y' = 5 \text{ if } x = 2$$

$$44 f''(x) = x^{1/3} - 5; \quad f'(1) = 2; \quad f(1) = -8$$

Exer. 45–46: Let $f(x) = 9 - x^2$ for $-2 \leq x \leq 3$, and let P be the regular partition of $[-2, 3]$ into five subintervals.

45 Find the Riemann sum R_P if f is evaluated at the midpoint of each subinterval of P .

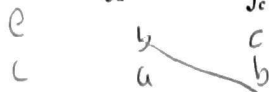
46 Find (a) A_{IP} and (b) A_{CP} .

Exer. 47–48: Verify the inequality without evaluating the integrals.

$$47 \int_0^1 x^2 dx \geq \int_0^1 x^3 dx \quad 48 \int_1^2 x^2 dx \leq \int_1^2 x^3 dx$$

Exer. 49–50: Express as one integral.

$$49 \int_c^e f(x) dx + \int_a^b f(x) dx - \int_c^b f(x) dx - \int_a^d f(x) dx$$



$$50 \int_a^d f(x) dx - \int_t^b f(x) dx - \int_a^g f(x) dx + \int_m^h f(x) dx + \int_1^a f(x) dx$$

51 A stone is thrown directly downward from a height of 900 feet with an initial velocity of 30 ft/sec.

(a) Determine the stone's distance above ground after t seconds.

(b) Find its velocity after 5 seconds.

(c) Determine when it strikes the ground.

52 Given $\int_1^4 (x^2 + 2x - 5) dx$, find

(a) a number z that satisfies the conclusion of the mean value theorem for integrals (5.28)

(b) the average value of $x^2 + 2x - 5$ on $[1, 4]$

[c] 53 Evaluate $\int_0^{10} \sqrt{1 + x^4} dx$ by using

(a) the trapezoidal rule, with $n = 5$

(b) Simpson's rule, with $n = 8$

(Use approximations to four decimal places for $f(x_k)$, and round off answers to two decimal places.)

[c] 54 To monitor the thermal pollution of a river, a biologist takes hourly temperature readings (in $^{\circ}\text{F}$) from 9 A.M. to 5 P.M. The results are shown in the following table.

Time of day	9	10	11	12	1
Temperature	75.3	77.0	83.1	84.8	86.5

Time of day	2	3	4	5
Temperature	86.4	81.1	78.6	75.1

Use Simpson's rule and Definition (5.29) to estimate the average water temperature between 9 A.M. and 5 P.M.